# Enhance the bike-sharing demand prediction app with city details plots

Estimated time needed: **90** minutes

In the previous lab, you have built a R Shiny app with leaflet to show the max bike-sharing demand predictions for each city. In the app, users can also select a specific city to show its detailed weather label.

Next, you are asked to use ggplot to render some more detailed plots such as bike-sharing prediction trend, temperature trend, humidity and bike-sharing demand prediction correlation, when users zoom-in to a city.

This lab includes the following tasks:

- **TASK**: Add a static temperature trend line
- **TASK**: Add an interactive bike-sharing demand prediction trend line
- **TASK**: Add a static humidity and bike-sharing demand prediction correlation plot

Let's start by opening RStudio project on IBM Watson Studio you created in the previous lab.

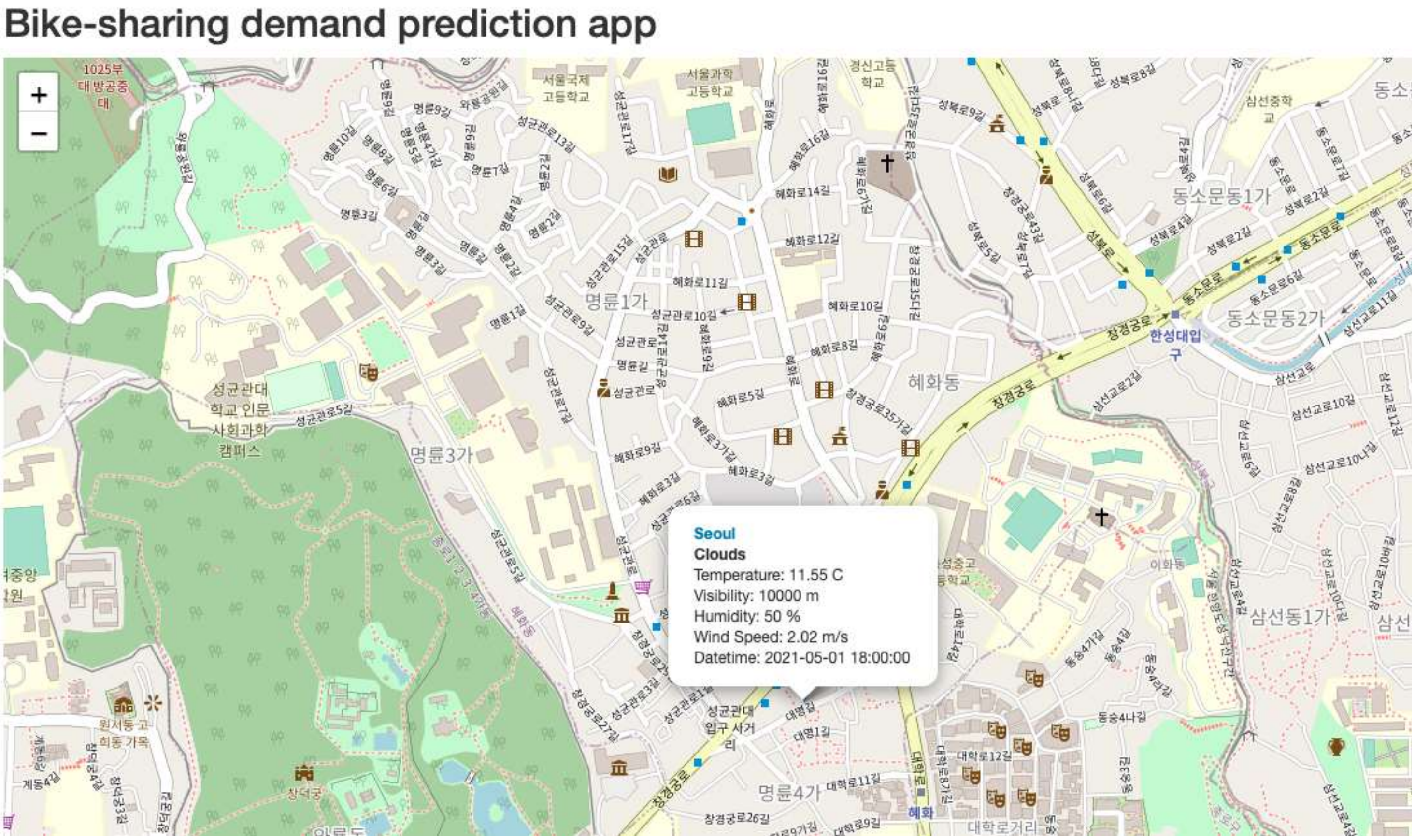
# Task 1: Add a static temperature trend line

When users select a city to zoom-in, one important information they may want to know is the temperature trend in the next 5-days. Such trend information can be visualized using line chart.

- Open `ui.R`, add a plot output with in `sidebarPanel()` with id `temp_line` and any other options you want to add such as height and width.
- Open `server.R`, locate to the `if-else` code branch checking if All cities or a specific city is selected in the `observeEvent()` function, then add a temperature trend plot using `renderPlot(...)` function with following configurations:
  - Add a `geom_line` to show the trend line
  - Add a `geom_point` to show temperature point
  - Add a `geom_text` to show the value on the plot

Note that the data for the trend line should come directly from `city_weather_bike_df`, instead of the `cities_max_bike` data frame used for leaflet.

Your temperature trend line should look similar to the following example




From this plot, users can easily check the temperature trend for the selected city.

# Task 2: Add an interactive bike-sharing demand prediction trend line

Another important trend line for a selected city is its bike-sharing demand prediction trend. It will also be a line chart but let's try to add some interaction to the chart so that users can click the point and see its bike-sharing demand and datetime.

- Open `ui.R`, add a plot output with in `sidebarPanel()` with id `bike_line` and a click event `click = "plot_click"`. You may add other options you want such as height and width.
- In `ui.R`, also add a `verbatimTextOutput("bike_date_output")` text output with id `bike_date_output`. So that the point you click on the plot can be shown in here.
- Open `server.R`, locate to the `if-else` code branch checking if all cities or a specific city is selected in the `observeEvent()` function, then add a bike-sharing demand prediction trend plot using `renderPlot(...)` function with following configurations:
  - Add a `geom_line` to show bike-sharing prediction trend line with the `x` axis represents the `FORECASTDATETIME` and the `y` axis represents `BIKE_PREDICTION`
  - Add a `geom_point` to show bike-sharing
  - Add a `geom_text` to show the value on the plot

Users then should able to click on the trend line to show the bike-sharing prediction and datetime values so let's add a text output rendering here.

- In `server.R` file, add a `renderText()` function to create a formatted text output showing the clicked x and y value.

```
output$bike_date_output <- renderText({
       ...
      })

```

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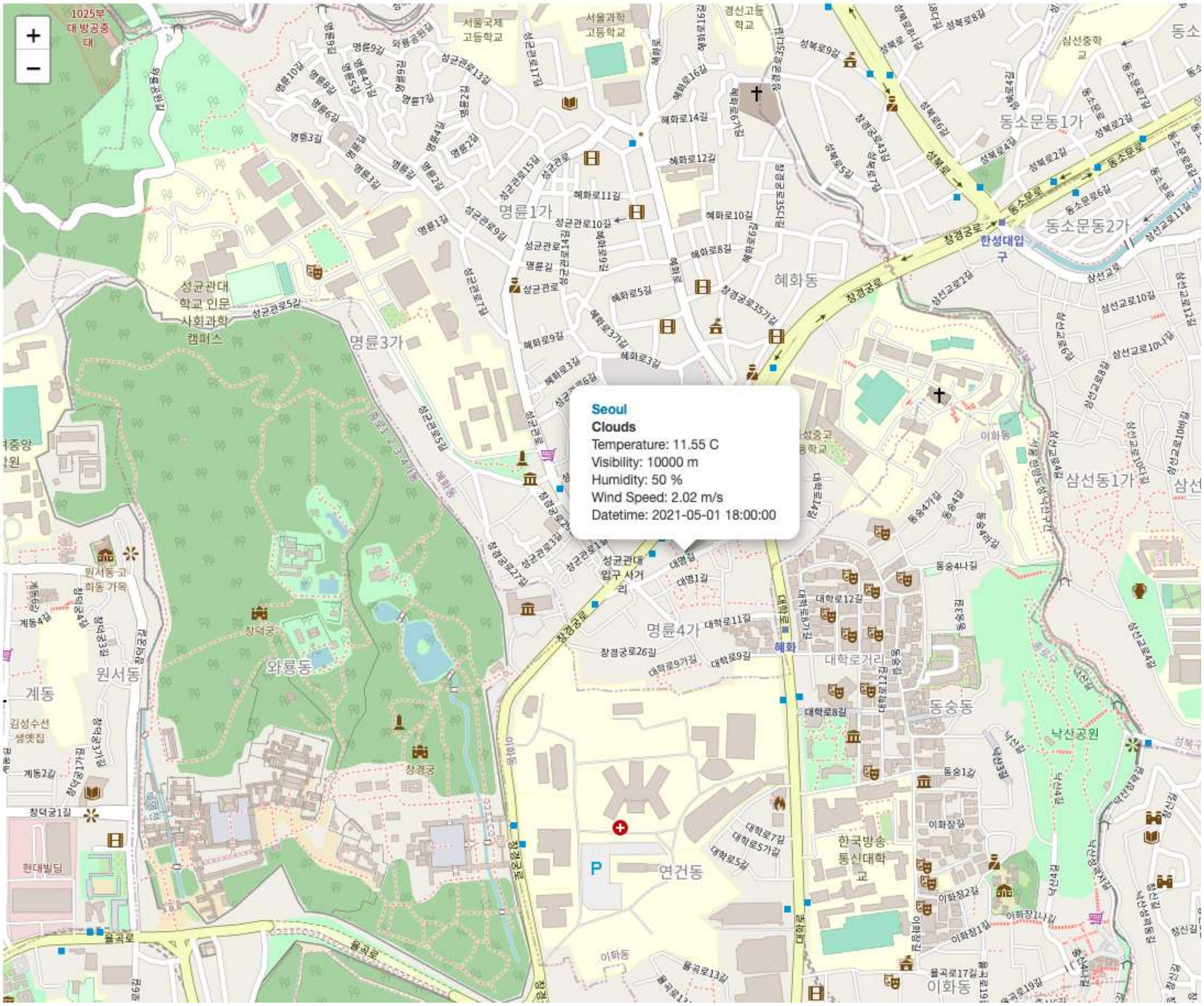
Your bike-sharing demand prediction trend plot should look similar to the following example:

Bike-sharing demand prediction app



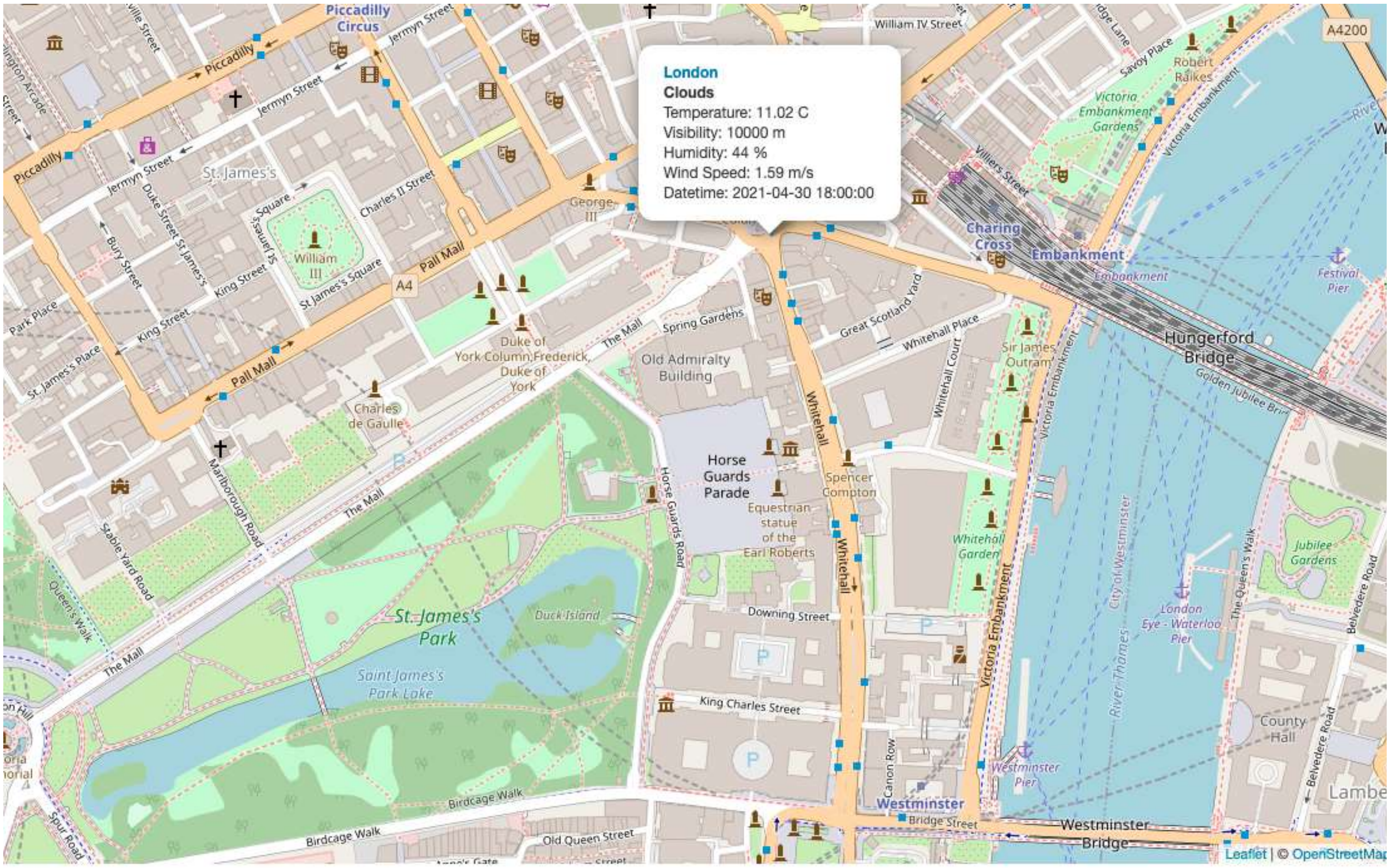

From this trend line, users can easily see the bike-demand prediction trend in the next 5-days. Users can also click the points on the plot to show its detailed prediction and datetime value.

# Task 3: Add a static humidity and bike-sharing demand prediction correlation plot

From the previous analysis, we know the bike-sharing demand value highly depends on the humidity value. So it would be great to show such correlation scatter point plot when a city is selected.

- Open `ui.R`, add a plot output with in `sidebarPanel()` with id `humidity_pred_chart`. You may add other options you want such as height and width.
- Open `server.R`, locate to the `if-else` code branch checking if all cities or a specific city is selected in the `observeEvent()` function, then add a trend plot using `renderPlot(...)` function with following configurations:
  - Add a `geom_point()` with x axis represents `HUMIDITY` and y axis represents `BIKE_PREDICTION`
  - Add a `geom_smooth()` with method `lm` and formula `y ~ poly(x, 4)`. You may choose any high order polynomial terms here

Your correlation plot should look similar to the following example:




# Next steps

Congratulations! You are almost finished the entire capstone project. Next, you will be preparing how to present your work and data-driven insights using PowerPoint.

## Author(s)

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## Change Log

| Date (YYYY-MM-DD) | Version | Changed By | Change Description |
|---|---|---|---|
| 2021-04-28 | 1.0 | Yan | Created the initial version |
| 2023-05-08 | 1.1 | Eric Hao | Updated Page Frames |
| 2023-05-08 | 1.2 | Jaskomal Natt | Updated markdown |